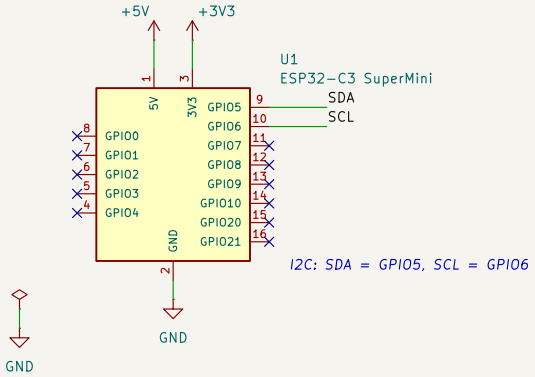
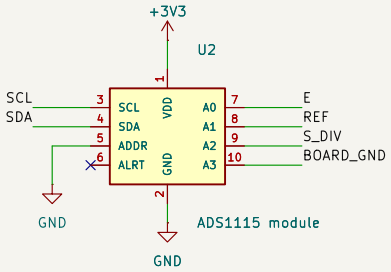
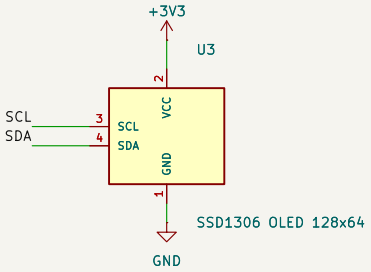
pH electrode simulator and leakage tester

1. Power the ADS1115 and the OLED from 3.3 V, never from 5 V.
2. The simulator voltage E is measured BEFORE R2; the board input VIN is worked out from S.
3. Leakage: I = (VIN − E) / R2 after a two-point calibration with SW2 pressed.
4. The simulator floats: its reference REF is tied only to VMID (R6, R7) and to the BNC shield.

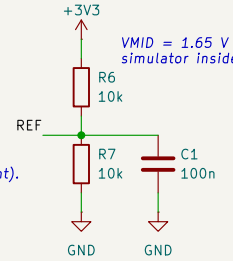




ADDR to GND: address 0x48
A0 – A1: simulator voltage E
A2: S / 2
A3: voltage on the shunt R5



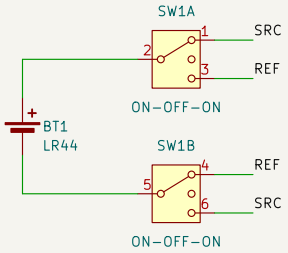
Address 0x3C. Wire by the printed pin names.



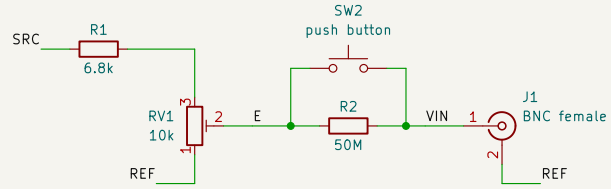
VMID = 1.65 V (R6, R7, C1) holds the floating simulator inside the ADS1115 input range.

J2 to P2 of the pH board, straight cable:
1 +5V 2 GND (through R5) 3 S

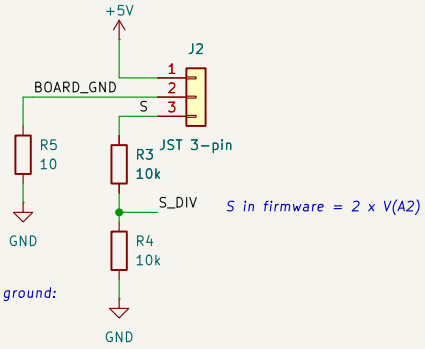
SW2 pressed: R2 shorted, the board input is E (calibration).
SW2 released: the signal goes through R2 (leakage measurement).



SW1 is cross-wired: the two ON positions give +E and −E, the middle one switches it off.



Never connect the ADS1115 to VIN: its input would become the leakage.



S in firmware = 2 x V(A2)

R5 = Rsh, shunt in the board ground:
board current = V(A3) / R5

Hardware licence: CERN-OHL-S-2.0
Wiring as in docs/wiring.png; firmware src/main.cpp.
igorlab

Sheet: /
File: calibrator.kicad_sch

Title: pH electrode simulator and leakage tester

Size: A4	Date: 2026-10-21	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 1/1